

Vamshi Kalyan Thummala

Overland Park, Kansas | +1 913 278 1319 | thvamshi02@gmail.com | [linkedin.com/in/vamshi-t02](https://www.linkedin.com/in/vamshi-t02)

PROFESSIONAL SUMMARY

- Data Engineer with 5+ years of experience across Data Engineering and Data Analytics, delivering scalable and reliable data platforms for enterprise environments.
 - Strong expertise in SQL, Python, data cleansing, and exploratory analysis, transforming raw data into high-quality, analytics-ready datasets.
 - Proven experience designing and optimizing ETL/ELT pipelines and data integration, transformation, and orchestration frameworks.
 - Hands-on experience building cloud data ecosystems (AWS, Azure, GCP) supporting modern data warehouses and analytics workloads.
 - Skilled in designing OLTP/OLAP models and dimensional schemas to enable efficient analytical querying and reporting.
 - Extensive background in KPI development, BI reporting, BI dashboards, and reporting solutions supporting Business Intelligence initiatives.
 - Experience delivering healthcare analytics solutions aligned with data governance, quality, and compliance standards.
 - Applied statistical and analytical techniques to support data-driven decision-making and advanced analytics use cases.
 - Implemented data governance and quality frameworks to ensure trusted, consistent, and secure enterprise data assets.
 - Collaborative problem-solving professional who partners with technical and business teams to deliver robust data solutions, promote a self-service analytics culture, and drive continuous analytical innovation.
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TECHNICAL SKILLS

- **Programming & Querying** - Python (NumPy, Pandas), SQL, Java, R, Shell Scripting
 - **Data Engineering & ETL** - Apache Airflow, DBT, Spark / PySpark, Scala, SSIS, Informatica, DataStage, System Integration, ETL/ELT Pipelines, Data Modeling, XML, JSON, Parquet, Workflow Orchestration, Debugging
 - **Databases & Data Warehousing** - Relational Databases (PostgreSQL, MySQL, MS SQLServer), Snowflake, NoSQL, Hive, TSQL, Teradata, MongoDB, OLTP/OLAP, Dimensional Modeling, Stored Procedures, Query Optimization
 - **Big Data & Cloud Platforms** - Hadoop, AWS (S3, Lambda, EC2, RDS), Azure, GCP, Data Lakes, Data Warehouses, Serverless Data Processing
 - **Analytics & Business Intelligence** - Power BI, Tableau, KPI Reporting, Exploratory Data Analysis (EDA), Advanced Excel, Dataflows Gen2, Dashboarding, Business Insights
 - **DevOps & Automation** - Git/GitHub, Jenkins, Docker, Kubernetes, CI/CD Pipelines, AWS CodePipeline, Jira
 - **Data Governance & Quality** - Data Governance, Data Validation, Metadata Management, Data Quality Frameworks
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PROFESSIONAL EXPERIENCE

Cigna Healthcare
Data Engineer

June 2025 - Present
Plano, Texas

- Developed enterprise data pipelines using Python, Spark, and Airflow to process millions of healthcare records daily, reducing ETL runtime by 40% and enhancing data availability for analytics and clinical teams.
- Designed and implemented RAG-enabled data pipelines leveraging OpenAI APIs, LangChain, and Hugging Face, enabling advanced information retrieval and NLP-driven insights from unstructured clinical datasets.
- Built predictive analytics and statistical modeling solutions using Python and R, including regression and segmentation models, improving patient risk stratification and claims forecasting accuracy by 30%.
- Developed curated data marts for claims analytics and clinical insights, enabling faster querying in Snowflake and supporting HIPAA-compliant reporting dashboards for stakeholders.
- Implemented feature engineering pipelines for machine learning models using MLflow and Spark, standardizing PHI masking and data transformation processes for HIPAA-compliant datasets.

- Leveraged Vertex AI, FAISS, and Hugging Face embeddings to build NLP-based recommendation and classification systems, accelerating clinical knowledge retrieval and decision support for healthcare teams.
- Designed and maintained scalable Medallion architecture pipelines, implementing bronze, silver, and gold layers with robust schema evolution across multiple data domains.
- Collaborated with data science teams to optimize segmentation datasets and regression models, resulting in actionable clinical insights and improved patient engagement strategies.
- Established robust data security, governance, and explainability frameworks, ensuring all data pipelines and analytics comply with HIPAA regulations while maintaining model interpretability.
- Automated ETL workflows and monitoring using Airflow and Spark, reducing data inconsistencies by 35% and ensuring real-time availability of high-quality datasets for predictive modeling.
- Supported data warehouse application management and analytic products, assisting with data security, change management, database support, warehouse implementation, data acquisition, and archive recovery within the enterprise data ecosystem.
- Managed and developed data integration processes, fulfilled data integration requests, handled new data sources, tested and resolved integration issues, and ensured proper user access controls across databases.
- Collaborated with Data Engineers, CDO Office, project and business leaders to scope, plan, and prioritize work, manage data sourcing, migration, quality, design, and assess impacts of database changes.
- Designed PHI masking strategies and HIPAA-compliant data pipelines, safeguarding sensitive patient information while enabling advanced analytics and ML experimentation.
- Integrated GenAI solutions with enterprise data pipelines to provide context-aware clinical and claims insights, leveraging LangChain and OpenAI APIs for dynamic query generation and summarization.
- Collaborated cross-functionally to deploy CI/CD pipelines, perform code reviews and automated model tracking via MLflow, ensuring reproducibility, scalability, and continuous improvement of healthcare analytics workflows.

Codal

Data Engineer

Nov 2022 – Dec 2024

India

- Engineered end-to-end ETL pipelines using Python, SQL, Airflow, and Fivetran to extract, transform, and load data from APIs, databases, and cloud storage into Redshift and BigQuery, enabling scalable analytics for business stakeholders.
- Designed and implemented Infrastructure-as-Code (IaC) solutions using Terraform on AWS and GCP, provisioning and managing cloud resources for high-availability ETL and real-time stream processing workflows.
- Developed real-time data pipelines leveraging Apache Spark, Kafka, Flink, and AWS Kinesis, enabling near-instant insights for operational analytics and monitoring high-volume streaming data.
- Built OLTP and OLAP architectures with Dimensions and Facts modeling in Redshift and BigQuery, improving query performance by 60% and supporting complex business intelligence reporting.
- Implemented CI/CD pipelines for ETL jobs and microservices using Jenkins and Git, automating deployment, reducing release cycles, and ensuring consistent performance across multiple environments.
- Developed serverless data workflows using AWS Lambda, Glue, and Airflow to orchestrate complex ETL processes, improving operational efficiency and reducing infrastructure costs by 25%.
- Established robust data governance, security, and validation frameworks, integrating automated data quality checks and root cause analysis to maintain data accuracy, compliance, and regulatory adherence.
- Created RESTful APIs and microservices for data ingestion and retrieval, enabling cross-functional teams to access high-quality, real-time datasets while ensuring secure and scalable integration.
- Monitored and optimized cloud infrastructure and ETL performance using AWS CloudWatch, Spark tuning, and query optimization, enhancing system scalability and reducing computational costs by 30%.
- Collaborated in Agile/Scrum teams to design, develop, and deploy high-performance data pipelines and analytics solutions, ensuring alignment with business priorities and continuous improvement of data operations.

Simform

Data Analyst

May 2021 – Nov 2022

India

- Performed exploratory data analysis and statistical modeling on large datasets using SQL and Python (Pandas, NumPy), identifying trends and patterns that informed product strategy and optimized business processes.
- Built automated reporting solutions and KPI dashboards using Power BI, Tableau, and Looker, providing real-time visibility into business metrics and reducing manual reporting efforts by 70%.

- Designed, implemented, and maintained ETL pipelines using SSIS, T-SQL, and Python to integrate data from CRM systems, databases, and third-party APIs, ensuring consistent, accurate, and timely data availability.
- Conducted root cause analysis and quantitative analysis on operational and product data, providing actionable insights that improved decision-making and enhanced executive-level strategic planning.
- Developed statistical models, A/B testing frameworks, and experimentation pipelines, resulting in a 25% improvement in user engagement and informed data-driven product optimizations.
- Created and maintained technical documentation, version control repositories, and data dictionaries, improving transparency, collaboration, and reproducibility across cross-functional teams.
- Applied advanced Excel techniques including VLOOKUP, Pivot Tables, and complex formulas to support ad-hoc analysis, reporting, and KPI monitoring for business stakeholders.
- Leveraged SSRS and SSAS to create standardized reports and OLAP cubes for multidimensional analysis, enabling efficient querying and deep business insights for leadership teams.
- Collaborated with engineering, product, and analytics teams to define data requirements, implement tracking mechanisms, and establish data governance standards, improving data completeness and accuracy by 45%.
- Utilized critical thinking, leadership, attention to detail and communication skills to present findings to executives and cross-functional teams, influencing strategic decisions and promoting a data-driven culture.

CERIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate

Jan 2026

Deep Learning Specialization – DeepLearning.AI (Coursera)

Aug 2020

EDUCATION

University of Central Missouri
Master's in Computer Science

Jan 2024 – June 2025
Warrensburg, Missouri

Keshav Memorial Institute of Technology
Bachelor of Technology in Information Technology

Aug 2019 – May 2023
Hyderabad, India